

Software Requirements Engineering

Project Documentation

Job Portal for Fresh Graduates

1. System Overview

1.1 Description

The Job Portal for Fresh Graduates is a web-based platform that connects final year university students and recent graduates with employers looking to hire entry-level candidates. Students build digital profiles and upload resumes. Employers post job listings. An admin moderates all listings before they go live. An AI keyword matching engine then compares student profile keywords against job descriptions and generates a match percentage visible to both parties.

1.2 Problem Statement

University placement processes in Pakistan are largely manual. Job opportunities reach students through WhatsApp groups, printed notices, or emails often days after the posting. Students have no way to track where their application stands. Employers receive unfiltered CVs and spend significant time screening irrelevant candidates. The placement office manually coordinates between both sides for every single opening, which is slow and error-prone.

1.3 Purpose

This system is built to digitize and automate the entire placement process from job posting to candidate matching to application tracking in one platform. The goal is to reduce placement time and help the right candidates reach the right employers faster.

2. Initiation Phase

2.1 Feasibility Study

The feasibility study determines whether the project should proceed. It is conducted before any development begins and evaluates the system across five dimensions.

2.1.1 Technical Feasibility

The system requires a web frontend, a backend with a database, an AI matching module, file storage for resumes, and an email notification service. The table below checks whether each component is achievable with available technology.

Component	Solution	Status
Frontend	HTML, CSS, React	Available

Backend	Node.js with MySQL	Available
AI Keyword Matching	Python spaCy library	free and open source
Resume File Storage	Server storage or AWS S3	Available
Email Notifications	SMTP or SendGrid API	Available free tier sufficient
Team Skill Match	Web development and basic NLP	Confirmed

No experimental or unavailable technology is required.

Verdict: Technically Feasible

2.1.2 Economic Feasibility

The system is economically feasible because it uses free and open-source technologies. Only minimal cost is required for hosting, storage, and deployment, which can be handled within a low student budget.

The university currently spends on annual job fairs, printed materials, and manual staff coordination. This portal replaces those recurring costs after the first year. Faster placements also improve university rankings, which carries long-term institutional value. The investment is justified.

Verdict: Economically Feasible

2.1.3 Operational Feasibility

User Group	Assessment	
Students	Already use digital platforms so minimal training needed	
Employers	Familiar with platforms like LinkedIn, a low learning curve.	
Admin Staff	Simple approve/reject workflow, 1 to 2 days training	
Placement Officer	Motivated to adopt replaces slow manual process	

All user groups have the capability and motivation to use this system.

Verdict: Operationally Feasible

2.1.4 Schedule Feasibility

The Job Portal for Fresh Graduates is schedule feasible because it can be completed within one semester. The project has limited modules, so the work can be done step by step, including design, development, testing, and final documentation. Therefore, the timeline is realistic for a semester project.

The project deadline is 6 months (24 weeks). The estimated timeline of 22 weeks keeps a 2-week buffer for unexpected delays.

Verdict: Schedule Feasible

2.1.5 Legal Feasibility

Legal Concern	Requirement
Student personal data	Secure storage and documented privacy policy required
Employer verification	Mandatory before account activation to prevent scam listings
Data sharing	Not permitted without explicit user consent

No laws are violated as long as a privacy policy is in place and employer verification is enforced.

Verdict: Legally Feasible

2.2 Key Stakeholders

Stakeholder	Role	Primary Interest
Fresh Graduate Students	Primary User for Job Seeker	Find relevant jobs, apply, and track application status
Employers and HR Managers	Primary User for Job Poster	Post listings and find qualified candidates efficiently
University Placement Officer	System Operator	Approve listings, monitor activity, access reports
Admin Staff	Moderator	Verify employer accounts, handle disputes
University Management	Sponsor	Placement statistics, employer partnerships, reputation
IT Department	Technical Maintainer	System uptime, security, database backups

2.3 Initial Scope

In Scope

- Student registration, profile creation, and resume upload
- Employer registration with admin verification
- Job listing creation and admin approval workflow
- AI keyword matching with match percentage display
- Skill gap display per job listing
- Job search and filters by skill, industry, and location
- One-click job application with profile auto-attached

- Application status tracking: Applied → Under Review → Shortlisted → Rejected → Hired
- Job bookmark and saved jobs section
- In-portal messaging between employers and shortlisted candidates
- Email notifications for job matches, status updates, and approval outcomes
- Admin dashboard with placement statistics
- Automatic listing expiry after deadline

Out of Scope — Phase 1

- Mobile application for Android or iOS
- Internship and part-time job listings
- LinkedIn or external job board integration
- AI video interviews or automated screening
- Academic transcript management

2.4 Constraints

Constraints are fixed limitations that cannot be changed. All design decisions must operate within them.

ID	Type	Constraint
CON-01	Budget	The project is developed within a limited student budget using free or low-cost tools and services.
CON-02	Time	System should be completed within 6-month semester duration
CON-03	Technology	Must use university's existing Apache server and MySQL database
CON-04	Platform	Web only must support Chrome, Firefox, and Edge
CON-05	Security	HTTPS mandatory on all pages. passwords encrypted
CON-06	Privacy	Student contact details hidden from employers all contact via in-portal messaging
CON-07	Eligibility	Restricted to final-year students and graduates within 2 years of graduation
CON-08	AI limitation	AI-based job matching will be based only on keyword matching and not advanced machine learning models.

2.5 Assumptions

ID	Assumption	Risk if Incorrect
ASM-01	Every student has an active university email for registration	Students without university emails cannot verify their identity
ASM-02	Employers provide valid company registration details	Fraud accounts create scam listings that harm students
ASM-03	Students upload updated resumes in PDF format for consistent processing	AI matching scores become inaccurate or unreliable
ASM-04	Basic keyword matching is accurate enough for Phase 1	Results may be too weak and require a more complex model
ASM-05	The university student database is current and accurate	Wrong students may be approved or valid ones rejected
ASM-06	Admin approvals are processed within a reasonable time (e.g., 24 hours)	Time-sensitive jobs expire before students can apply
ASM-07	Stable internet is available on campus	System is inaccessible during outages
ASM-08	IT department provides server space and maintenance	No deployment environment without IT cooperation
ASM-09	Employers are willing to use this portal	No job listings posted portal becomes ineffective

3. Planning Phase

Research Methodology

Research was conducted to ensure that requirements are based on real user needs rather than assumptions.

Secondary Research

Existing job portals such as LinkedIn and Rozee.pk were reviewed to understand common features like job posting, profile creation, and application workflows. These observations helped in defining system requirements.

Primary Research

Informal interviews were conducted with final-year students to understand their job search experience. Most students reported difficulty in tracking applications and expressed interest in seeing job matching suggestions.

The university placement process was also observed, which showed that job postings are handled manually and take several days to reach students.

3.2 Requirements Elicitation

Requirements were gathered using simple techniques to understand user needs.

- **Interviews:** Conducted with placement staff and students to understand problems in job searching and posting.
- **Observation:** The university placement process was observed to identify delays and manual steps.
- **Document Analysis:** Existing job forms and online job portals were reviewed to define common system requirements.

3.3 Requirements Identification — MoSCoW Model

MUST HAVE

ID	Requirement	Type
FR-01	The system shall allow students to register using a university email and verify via a confirmation link	Functional
FR-02	The system shall allow students to create a profile containing name, degree, CGPA, skills, projects, and internship history	Functional
FR-03	The system shall allow students to upload a resume in PDF format not exceeding 5 MB	Functional
FR-04	The system shall allow employers to register with company name, registration number, and business email	Functional
FR-05	The system shall require admin to verify and approve employer accounts before activation	Functional
FR-06	The system shall allow verified employers to post listings with title, description, required skills, minimum CGPA, and application deadline	Functional
FR-07	The system shall allow students to search and filter job listings by skill, industry, and location	Functional
FR-08	The system shall calculate a match score based on keyword similarity between resumes and job descriptions.	Functional
FR-09	The system shall track application status through: Applied, Under Review, Shortlisted, Rejected, Hired	Functional

NFR-01	All passwords shall be encrypted using bcrypt before storage	Non-Functional
NFR-02	The system shall enforce HTTPS on all pages and API calls	Non-Functional
NFR-03	The system shall work on the latest versions of Chrome, Firefox, and Edge	Non-Functional

SHOULD HAVE

ID	Requirement	Type
FR-10	The system shall display a skill gap section per job listing showing matched and missing skills	Functional
FR-11	The system shall allow students to bookmark listings and access them from a Saved Jobs section	Functional
FR-12	The system shall notify students by email when a job matching at least 3 of their keywords is posted	Functional
FR-13	The system shall provide in-portal messaging between employers and shortlisted candidates	Functional
FR-14	The system shall automatically close a listing after its application deadline passes	Functional
FR-15	The system shall allow employers to filter applicants by CGPA range and specific skill keywords	Functional
NFR-04	All pages shall load within 3 seconds on a standard 4G connection	Non-Functional
NFR-05	The system shall support 500 concurrent users without performance degradation	Non-Functional

COULD HAVE

ID	Requirement	Type
FR-16	The system shall show a profile completion score guiding students to strengthen their profile	Functional
FR-17	The system shall allow employers to save shortlisted candidates for future job openings.	Functional
FR-18	Admin shall be able to export monthly placement reports as a PDF file	Functional

WON'T HAVE (Phase 1)

Explicitly excluded to protect timeline and budget.

ID	Feature	Reason
FR-19	Mobile application	Beyond Phase 1 scope, system is web-based only
FR-20	AI video interviews	Too complex for current scope
FR-21	LinkedIn profile import	Requires paid API agreement

3.4 Requirements Validation

After all requirements were documented a review was conducted to identify missing incomplete and inconsistent requirements before proceeding. This step helps ensure correctness and reduces the risk of errors in later development stages.

3.4.1 Missing Requirements

ID	What Was Missing	Fix Applied
VAL-M01	No way for a student to withdraw an application	Added: Students can withdraw applications before employer reviews them.
VAL-M02	No process defined when employer verification fails	Added: System notifies employer with rejection status.
VAL-M03	No account deletion or data removal requirement	Added: Users can request account deletion and data is removed within a defined period.
VAL-M04	No defined behavior when no suitable jobs are found	Added: System suggests updating skills or profile information.

3.4.2 Incomplete Requirements

ID	Affected Requirement	Problem	Fix Applied
VAL-I01	FR-08	Matching formula was not defined cannot be built without it	Added explicit formula: (Number of matching keywords ÷ Total job keywords) × 100
VAL-I02	FR-15	Trigger, recipient, and email content were not specified	Split into separate requirements with defined trigger and recipient

VAL-I03	FR-05	Verification steps and rejection process were not described	Added steps: admin checks company name, number, and email domain rejection requires written reason
----------------	-------	---	---

3.4.3 Inconsistent Requirements

ID	Conflict Found	Fix Applied
VAL-C01	FR-01 allowed all students to register but FR-02 implied final-year only	Clarified: only final-year students and graduates within 2 years are eligible
VAL-C02	FR-06 stated job listings are published immediately but FR-07 required admin approval	Clarified: Job listings remain in Pending status until approved by admin
VAL-C03	One requirement exposed student phone numbers while CON-08 required contact privacy	Fixed: Phone numbers hidden from employers all contact through in-portal messaging only
VAL-C04	NFR-05 set 500 user limit but no behaviour was defined under high application load	Added: Applications queued in order received — student sees loading indicator during delay

3.4.4 Validation Summary

Check	Issues Found	Issues Resolved
Missing Requirements	4	4
Incomplete Requirements	3	3
Inconsistent Requirements	4	4
Total	11	11

All eleven issues were resolved before finalising the requirements. The requirements baseline is approved and ready for the design phase.
